

José Manuel Robles-Aguilar

M.Sc. IN STATISTICAL PHYSICS · ARTIFICIAL INTELLIGENCE · PROTEIN SCIENCE · BIOPHYSICS

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Education

Cinvestav

M.Sc. IN PHYSICS

- GPA: 3.6

Mexico City, MX

01 Sep 2023 - 16 Dec 2025

Center for Computing Research (CIC)

SEMESTER

- Algorithm design and analysis, and quantum computing courses

Mexico City, MX

01 Sep 2022 - 01 Jan 2023

National Polytechnic Institute (IPN)

BACHELOR DEGREE IN PHYSICS AND MATHEMATICS

- GPA: 3.4

Mexico City, MX

01 Sep 2017 - 01 Jul 2022

Research Experience

MSc Thesis: "AI-driven Prediction of Circular Dichroism Spectra of Proteins"

CINVESTAV - DEPT OF PHYSICS

- Advisors: Dr. Mauricio Demetrio Carbajal-Tinoco, Dr. Damián Jacinto-Méndez
- Implemented deep Bayesian neural networks (BNNs) into the KCD model increasing the accuracy by 40%

Mexico City, MX

Nov 2024 - Nov 2025

AI+Science Summer School

UNIVERSITY OF CHICAGO JOHN W. BOYER CENTER

- Organizer: University of Chicago Data Science Institute

Paris, FR

Jul 2025

Fluorescence Microscopy project: "Brownian-particle motion inside lipid vesicles"

SOFT MATTER LABORATORY - CINVESTAV - DEPT OF PHYSICS

- Advisor: Dr. Damián Jacinto-Méndez

Mexico City, MX

May 2024 - Jul 2024

Proteomics Project: "Radius of Gyration in Alpha-helix and Beta-sheet proteins"

CINVESTAV - DEPT OF PHYSICS

- Advisor: Dr. Mauricio Demetrio Carbajal-Tinoco

Mexico City, MX

Jan 2024 - Feb 2024

International Research School

DURHAM UNIVERSITY

- Organizers: Durham University, Queen Mary University of London, University of Leeds

Durham, UK

Jan 2022

Teaching Assistantship

CENTER FOR COMPUTING RESEARCH (CIC)

- Advisor: Dr. Sun GuoHua

Mexico City, MX

Sep 2021 - Mar 2022

Institutional Incentive Grant for Research Training (BEIFI)

SUPERIOR SCHOOL OF PHYSICS AND MATHEMATICS (ESFM-IPN)

- Advisors: Dr. Gerardo Ortega-Cervantes, Dr. Ana María Paniagua-Mercado and Dr. Norma Sanchez-Salas

Mexico City, MX

2018 - 2022

Poster Presentations

Robles-Aguilar, J. M., Carbajal-Tinoco, M. D., Jacinto-Méndez, D. 2025. *AI-driven Prediction of Circular Dichroism Spectra of Proteins*. 13th Symposium on Structural Proteomics, Human Technopole, Milan, Italy.

Robles-Aguilar, J. M., Carbajal-Tinoco, M. D., Jacinto-Méndez, D. 2025. *The knowledge-based Circular Dichroism (KCD) method*. Physics and Life Days Conference, Institut Jacques Monod, Paris, France.

Skills

Deep Learning, Machine Learning, GenAI, Prompt Engineering, Ubuntu, Python, PyTorch, SciKit-learn, Data Analysis, Microsoft Office, Excel, PowerBI, Matlab, Statistics, Structural Biology, Proteomics, Biophysics, Bioinformatics, Quantum Computing, Fluorescence Microscopy, XRD, AFM, EM, UV-Vis, Raman. Languages: Spanish (Native), English (C1), French (B1).

Ongoing Projects and Learning

Project: Graph neural networks (GNNs) for predicting binding affinity between a drug molecule and a target protein.

Learning: Drug discovery, Molecular Dynamics, R, SQL.